

20 Power Query {M}agic Tips Self- Service Developer Should Know

Cloud Data Driven
Oct-2025

Presented by
David Kofod Hanna



twoday

You just have a self-
service solution...

David Kofod Hanna



Senior Advisor, Data Storytelling @ twoday

+200 courses as Academy Trainer and 10 years as consultant
MCT, Certified Tabular Editor 3 Trainer, DP-600, DP-700, PL-300, CPUX-F



Passionate about guiding self-service Power BI

developers for more enterprise manageable concepts in a consumable and practical way



Born on beautiful “Sunshine island”: Bornholm

Lives in Silkeborg with wife and 3 kids
Love football and running half-marathons

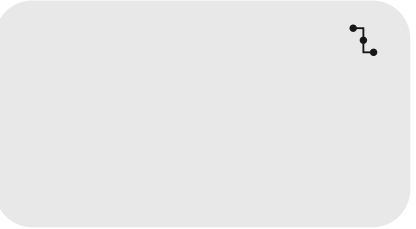
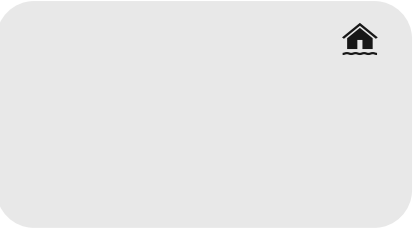
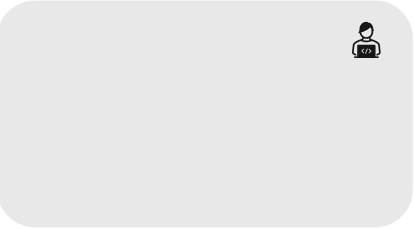
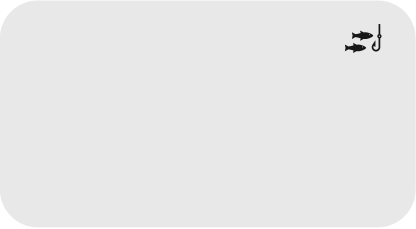
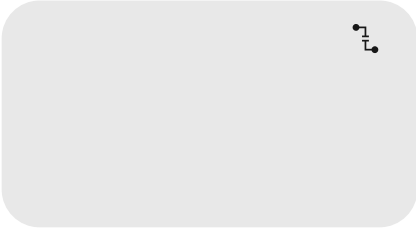
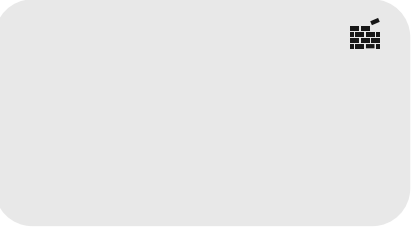
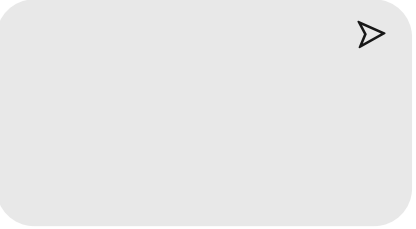
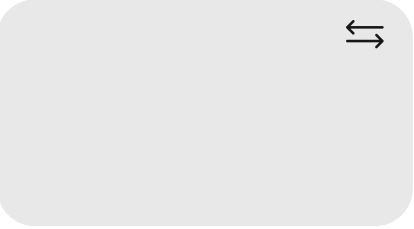
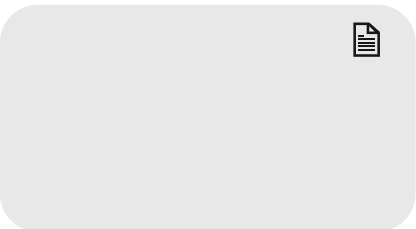
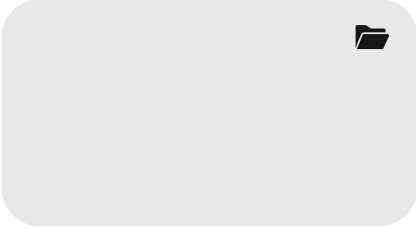
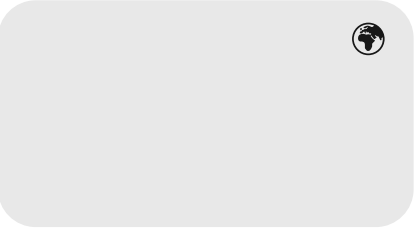
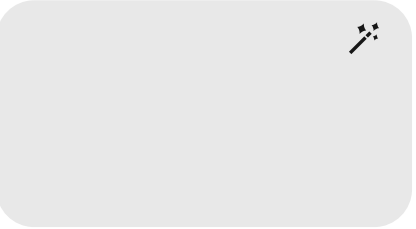
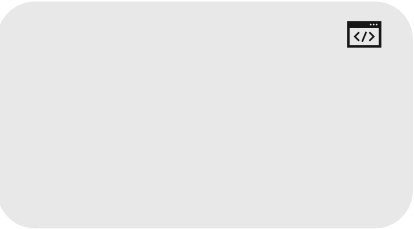
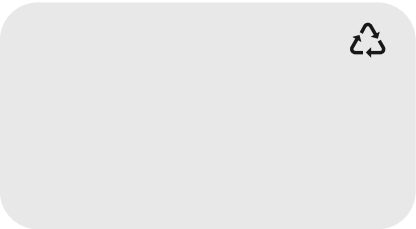
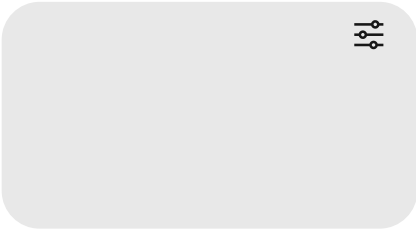
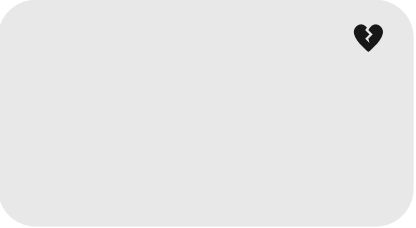
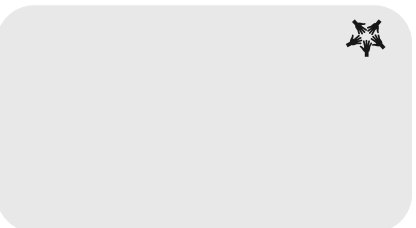
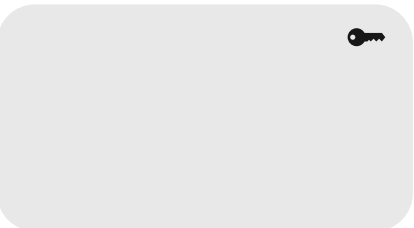
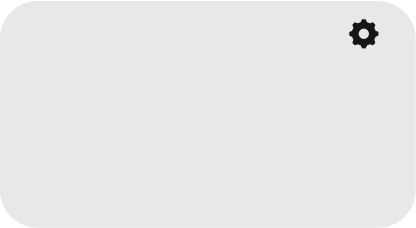
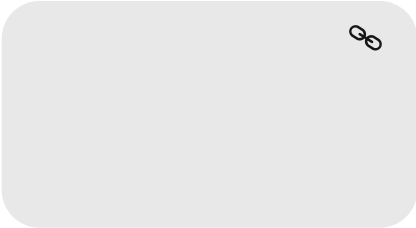


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How many prefer
Power Query & Excel
over
T-SQL, PySpark, Python?

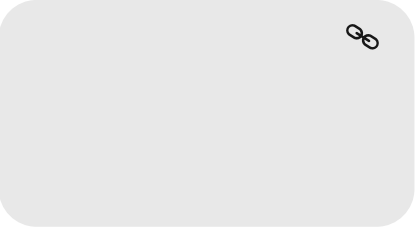
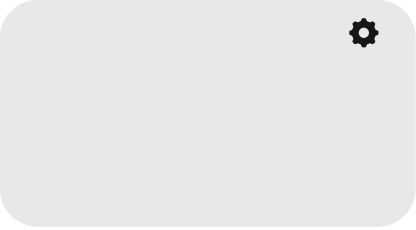
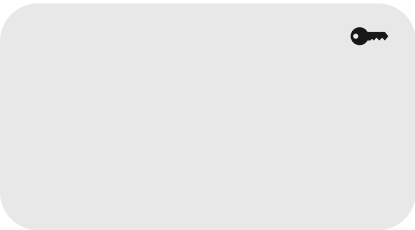
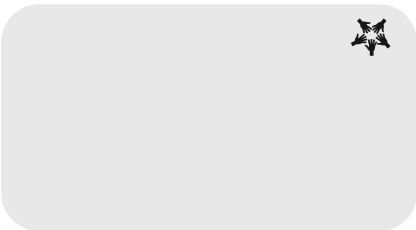
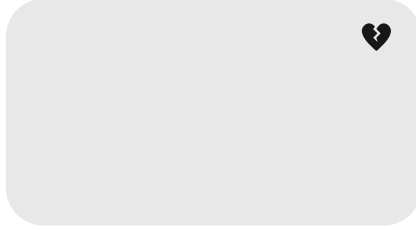
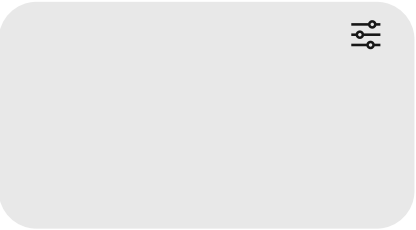
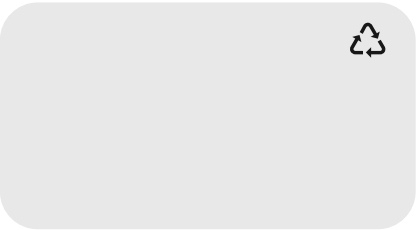
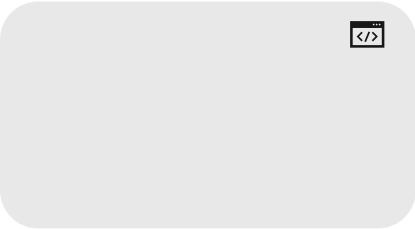
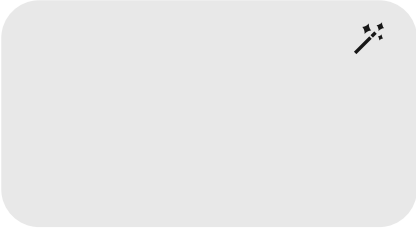
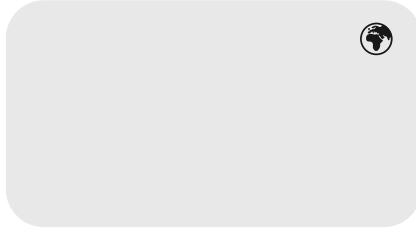
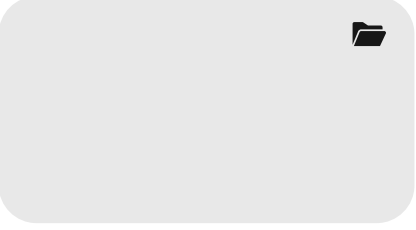
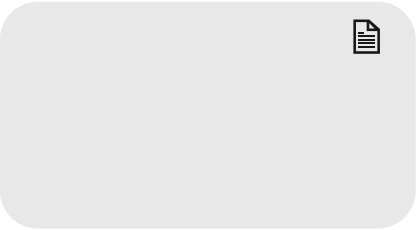


20 Power Query {M}agic Tips & Tricks



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 Power BI Desktop  Fabric only

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1.



Reference is preferred –
Medallion Architecture is that you?

2.



Choose Columns wisely –
Get the Gear Icon

3.



Don't create relationships
on attributes – create
surrogate keys

4.



Composite Key Merge
for multiple column join

5.



Red text will break you –
MissingField.Ignore or
UseNull

6.



Change settings with
autodetect column types for
unstructured sources

7.



TMDL Script for **Date** table

8.



**Don't use the Excel
Connector** when connecting
to Excel – use **Web**

9.



Unpivot – It feels like magic

10.



Stop Find and Replace for
regional formats by **Using
Locale...**

11.



Structure your **queries** in
folders

12.



Document **table properties**

13.



From Power Query to
Fabric Dataflows Gen 2 with
copy-paste

14.



**Dynamically Reference Your
Tables** for Table.Profile

15.



**Snapshots with Append to
Lakehouse**

16.



Don't Break the Fold – but if
you use Excel and .csv you
are!

17.



Look for the **stream** in View
Query Plan for optimization

18.



You can now **write SQL-
Views** with Visual Query
Editor

19.



From **Excel** to **SQL** Analytics
Endpoint in Lakehouse

20.



Orchestration Pipelines with
invoke Dataflow gen2 &
Semantic Model refresh

Relax – we have Microsoft to assist us...

Add data to your report
Once loaded, your data will appear in the **Data** pane.



Use sample data

Two ways to use sample data



Take a tutorial online

Learn how to create a robust report, step-by-step.

[Launch tutorial](#) 



Experiment on your own

To start creating visuals on your own, load sample data.

 **Load sample data**



Welcome to One Big Table – thank you Microsoft...

Table.TransformColumnTypes(financials_table,({ "Segment", type text}, {"Country", type text}, {"Product", type text}, {"Discount Band", type text}, {"Units Sold", type number}, {"Manufacturing Price", type number}, {"Sale Price", type number}, {"Sales Index", type number}, {"Discount", type number}, {"Sales", type number}, {"COGS", type number}, {"Profit", type number}, {"Date", type text}, {"Month Number", type number}, {"Month Name", type text}, {"Year", type number}))																
Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Sales Index	Discount	Sales	COGS	Profit	Date	Month Number	Month Name	Year	
1 Government	Canada	Cameras	None	2028.5	3	20	12270	0	32170	3835	12285	01/01/2014	1	January	2014	
2 Government	Germany	Cameras	None	1522	3	29	29400	0	25420	12219	12219	01/01/2014	1	January	2014	
3 Midmarket	France	Cameras	None	2178	3	35	32670	0	32670	10880	10880	01/06/2014	6	June	2014	
4 Midmarket	Germany	Cameras	None	888	4	15	18300	0	18300	4440	4440	01/06/2014	6	June	2014	
5 Midmarket	Mexico	Cameras	None	2478	4	15	47880	0	24780	12358	12358	01/06/2014	6	June	2014	
6 Government	Germany	Cameras	None	1243	3	268	92400	0	92400	38240	12628	01/12/2014	12	December	2014	
7 Midmarket	Germany	Monitors	None	301	3	15	18815	0	18815	4625	4625	01/01/2014	1	January	2014	
8 Channel Partners	Canada	Monitors	None	2218	3	12	30218	0	30218	7534	12562	01/06/2014	6	June	2014	
9 Government	France	Monitors	None	1889	3	20	17880	0	17880	38880	18880	01/06/2014	6	June	2014	
10 Channel Partners	Germany	Monitors	None	1545	3	12	38540	0	38540	4835	12803	01/06/2014	6	June	2014	
11 Midmarket	Mexico	Monitors	None	2470	3	15	37050	0	24700	12338	12338	01/06/2014	6	June	2014	
12 Enterprise	Canada	Monitors	None	2065.5	3	125	55137.5	0	109880	33527.5	109880	01/07/2014	7	July	2014	
13 Small Business	Mexico	Monitors	None	958	3	209	287400	0	287400	29930	47905	01/08/2014	8	August	2014	
14 Government	Germany	Monitors	None	2148	3	2	24602	0	10780	4260	10780	01/09/2014	9	September	2014	
15 Enterprise	Canada	Monitors	None	485	3	185	42175	0	42175	41460	1735	01/10/2013	10	October	2013	
16 Midmarket	United States of America	Monitors	None	425	3	15	6215	0	4250	4250	5575	01/12/2014	12	December	2014	
17 Government	Canada	Passes	None	262	18	20	1840	0	1840	2620	2620	01/02/2014	2	February	2014	
18 Midmarket	Mexico	Passes	None	374	18	15	24610	0	24610	3740	4878	01/02/2014	2	February	2014	
19 Channel Partners	Canada	Passes	None	2518	18	20	30218	0	30218	7534	12562	01/06/2014	6	June	2014	
20 Government	Germany	Passes	None	1088	18	20	281580	0	30548	30548	12014	01/06/2014	6	June	2014	
21 Channel Partners	Germany	Passes	None	587	18	20	1203	0	1203	1203	15014	01/07/2014	7	July	2014	
22 Government	Mexico	Passes	None	883	18	20	1551	0	4413	1788	1788	01/08/2014	8	August	2014	
23 Midmarket	France	Passes	None	349	18	20	6203	0	1480	2743	1480	01/09/2013	9	September	2013	
24 Small Business	Mexico	Passes	None	788	18	20	10440	0	18700	19440	19440	01/09/2013	9	September	2013	
25 Midmarket	Mexico	Passes	None	2472	18	20	17880	0	24720	12340	12340	01/09/2014	9	September	2014	
26 Government	United States of America	Passes	None	1149	18	20	8883	0	6713	2389	6713	01/10/2014	10	October	2014	
27 Government	Canada	Passes	None	1725	18	20	487700	0	448020	125218	125218	01/11/2013	11	November	2013	
28 Channel Partners	United States of America	Passes	None	412	18	20	2644	0	2758	4208	4208	01/11/2013	11	November	2013	
29 Midmarket	Canada	Passes	None	2152	18	20	11280	0	21520	10780	10780	01/12/2013	12	December	2013	
30 Government	Canada	Passes	None	1827	18	20	30540	0	38270	18270	18270	01/12/2014	12	December	2014	
31 Government	Germany	Passes	None	1515	18	20	528150	0	283380	126278	126278	01/12/2014	12	December	2014	
32 Government	Mexico	Passes	None	1893	120	20	33471	0	7403	2988	7403	01/01/2014	1	January	2014	
33 Enterprise	France	Passes	None	1894	120	20	12190	0	219480	3820	3820	01/01/2014	1	January	2014	
34 Channel Partners	Germany	Passes	None	2181	120	20	2181	0	2181	4481	4481	01/01/2014	1	January	2014	
35 Government	Germany	Passes	None	1088	120	20	952300	0	241380	86340	86340	01/06/2013	6	June	2013	
36 Channel Partners	Germany	Passes	None	1545	120	20	18480	0	38540	4835	12803	01/06/2013	6	June	2013	
37 Enterprise	United States of America	Passes	None	2421	120	20	352625	0	336320	14200	14200	01/08/2014	8	August	2014	
38 Enterprise	Canada	Passes	None	345	120	20	43125	0	43125	1721	1721	01/10/2013	10	October	2013	
39 Small Business	Canada	VTT	None	2061	200	200	800300	0	380219	380219	380219	01/02/2014	2	February	2014	
40 Channel Partners	Germany	VTT	None	2858	255	12	54058	0	54058	8514	25542	01/04/2014	4	April	2014	
41 Midmarket	France	VTT	None	2178	200	15	12670	0	12670	21780	12668	01/06/2014	6	June	2014	
42 Midmarket	Germany	VTT	None	988	200	15	18300	0	18300	9880	4440	01/06/2014	6	June	2014	
43 Government	France	VTT	None	2527	200	200	914400	0	394450	394450	127420	01/07/2013	7	July	2013	
44 Small Business	France	VTT	None	1181	200	388	443300	0	387330	387330	387330	01/09/2013	9	September	2013	
45 Government	Canada	VTT	None	1817	200	20	44480	0	44480	38170	18278	01/12/2013	12	December	2013	
46 Government	France	Antennas	None	2760	200	200	462400	0	462400	247980	247980	01/10/2014	10	October	2014	
47 Channel Partners	United States of America	Antennas	None	1863	200	12	24688	0	24688	3819	17877	01/04/2014	4	April	2014	
48 Enterprise	Germany	Antennas	None	4218.5	200	125	527437.5	0	527437.5	380340	21267.5	01/04/2014	4	April	2014	
49 Government	France	Antennas	None	1894	200	20	17680	0	17680	18940	18940	01/06/2014	6	June	2014	
50 Government	Germany	Antennas	None	1880	200	2	11802	0	11802	4430	8372	01/07/2014	7	July	2014	
51 Channel Partners	United States of America	Antennas	None	2142	200	12	25831	0	25831	9425	9425	01/08/2014	8	August	2014	
52 Government	United States of America	Antennas	None	1145	200	2	8001	0	8001	5713	2389	01/10/2014	10	October	2014	
53 Midmarket	United States of America	Antennas	None	825	200	15	8225	0	8225	6250	8875	01/12/2014	12	December	2014	
54 Government	France	Passes	None	8845	18	2	27618	0	27618	38729	38729	01/01/2014	1	January	2014	

Let's Star
Schema All the
things!

{1}

Reference is preferred –
*Medallion Architecture is
that you?*



{2}

Choose Columns wisely –
Get the **Gear Icon**



{3}

Don't create relationships on
attributes – **create surrogate
keys**



{4}

Composite Key Merge



{5}

Red text will break you



Red text will break you



✕ ✓ *fx*

= Table.SelectColumns(Source,{"Segment", "Product", "Discount Band"})

 Expression.Error: The column 'Discount Band' of the table wasn't found.
Details:
Discount Band

Option 1: Manual effort

✕ ✓ *fx*

= Table.SelectColumns(Source,{"Segment", "Product"})

	^B _C Segment	^B _C Product
1	Government	Carretera
2	Government	Carretera
3	Midmarket	Carretera
4	Midmarket	Carretera
5	Midmarket	Carretera

Red text will break you





= Table.SelectColumns(Source,{"Segment", "Product", "Discount Band"})

 Expression.Error: The column 'Discount Band' of the table wasn't found.
Details:
Discount Band

Option 2: Use MissingField.Ignore



= Table.SelectColumns(Source,{"Segment", "Product", "Discount Band"}, MissingField.Ignore)

	A ^B _C Segment	A ^B _C Product
1	Government	Carretera
2	Government	Carretera
3	Midmarket	Carretera
4	Midmarket	Carretera
5	Midmarket	Carretera
6	Government	Carretera
7	Midmarket	Montana
8	Channel Partners	Montana
9	Government	Montana

Red text will break you



✕ ✓ *fx*

= Table.SelectColumns(Source,{"Segment", "Product", "Discount Band"})

 Expression.Error: The column 'Discount Band' of the table wasn't found.
Details:
Discount Band

Option 3: Use MissingField.UseNull

✕ ✓ *fx*

= Table.SelectColumns(Source,{"Segment", "Product", "Discount Band"}, MissingField.UseNull)

	A^B_C Segment	A^B_C Product	? Discount Band
1	Government	Carretera	null
2	Government	Carretera	null
3	Midmarket	Carretera	null
4	Midmarket	Carretera	null
5	Midmarket	Carretera	null
6	Government	Carretera	null
7	Midmarket	Montana	null
8	Channel Partners	Montana	null
9	Government	Montana	null
10	Channel Partners	Montana	null
11	Midmarket	Montana	null

{6}

Change settings with
autodetect column types
for **unstructured** sources



Change settings with autodetect column types for **unstructured** sources

Settings in Power BI Desktop

GLOBAL

- Data Load
- Power Query Editor
- DirectQuery
- R scripting
- Python scripting
- Security
- Privacy
- Regional Settings
- Updates
- Usage Data
- Diagnostics
- Preview features
- Auto recovery
- Report settings
- Copilot (preview)

Type Detection

- ☐ Always detect column types and headers for unstructured sources
- ☒ Detect column types and headers for unstructured sources according to each file's setting
- ☐ Never detect column types and headers for unstructured sources

Background Data

- ☐ Always allow data previews to download in the background
- ☒ Allow data previews to download in the background according to each file's setting
- ☐ Never allow data previews to download in the background

Time intelligence

- ☐ Auto date/time for new files ⓘ [Learn more](#)

CURRENT FILE

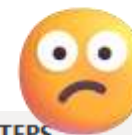
- Data Load
- Regional Settings
- Privacy
- Auto recovery
- Published semantic model settings
- Query reduction
- Report settings

Type Detection

- ☒ Detect column types and headers for unstructured sources

Time intelligence

- ☐ Auto date/time ⓘ [Learn more](#)



APPLIED STEPS	
Source	✱
Navigation	✱
Promoted Headers	✱
Changed Type	
Added Conditional Column	✱
Changed Type1	
Added Conditional Column1	✱
Added Conditional Column2	✱
Added Conditional Column3	✱
Changed Type2	
Inserted Sum	✱
Renamed Columns	
Merged Queries	✱
Expanded Accuracy Master	✱
Added Conditional Column4	✱
Changed Type3	
Filtered Rows	✱

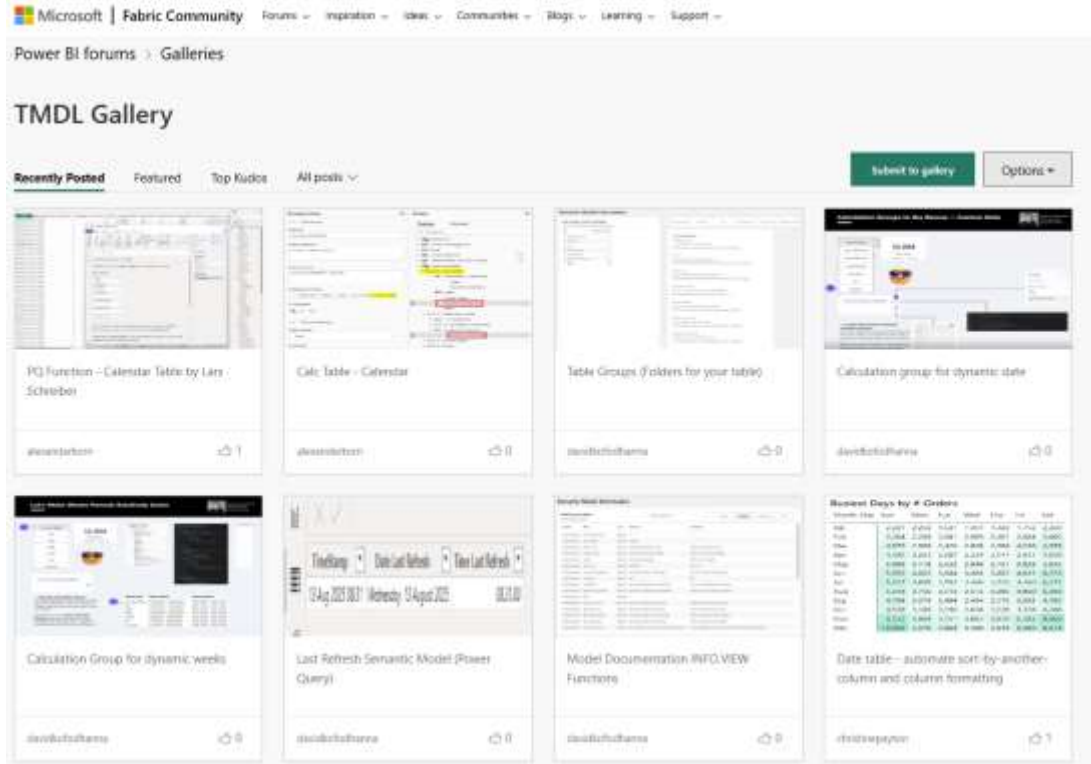
{7}

TMDL Script for Date table

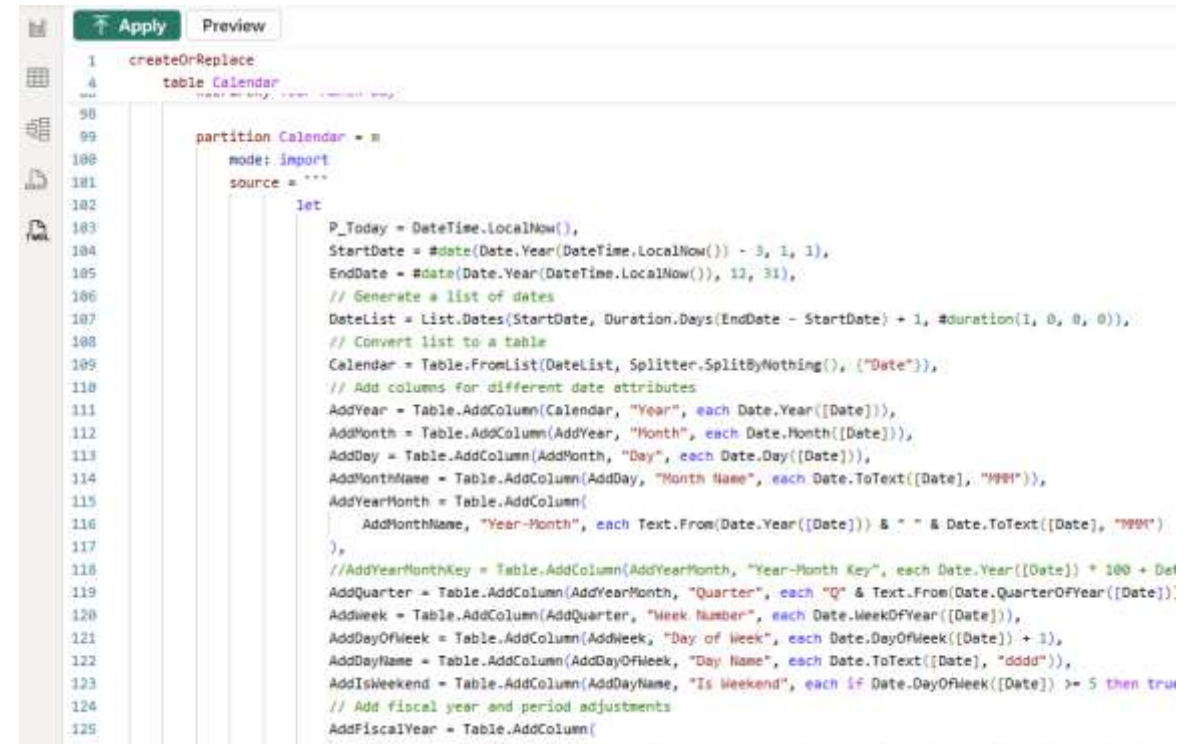


TMDL Script for Date table

TMDL Gallery



Paste, Edit, Run in TMDL in Power BI Desktop



{8}

**Don't use the Excel
Connector** when connecting
to Excel – use Web



{9}

Unpivot – It feels like magic



{10}

Stop Find and Replace
for regional formats by
Using **Locale...**



{11}

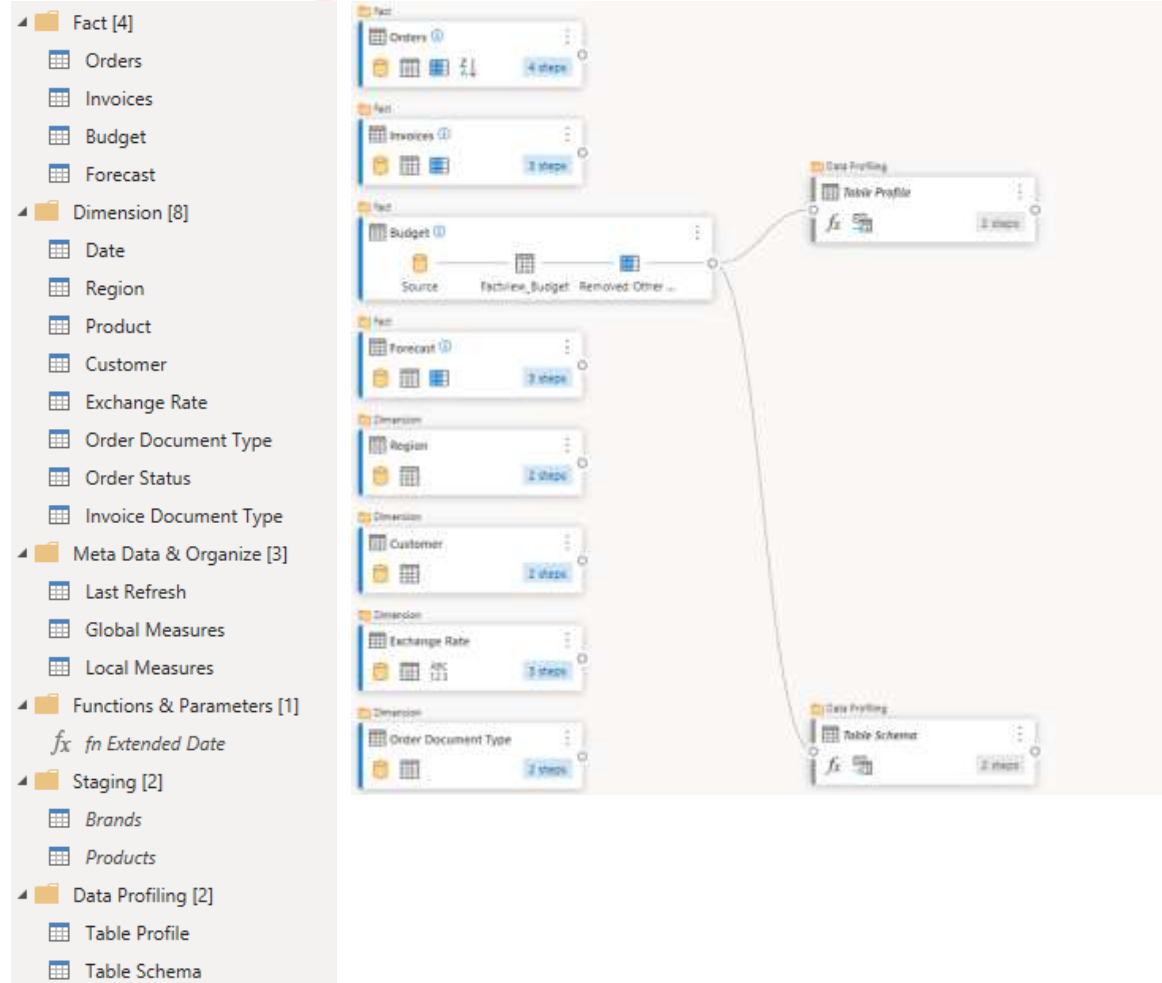
Structure your **queries** in
folders



Query folders by type – and Diagram View in Dataflow Gen2



Dimview Brands
Dimview Budget Rate
Dimview Customers
Dimview Employees
Dimview Exchange Rate
Dimview Invoice Document Type
Dimview Order Document Type
Dimview Order Status
Dimview Products
Dimview Regions
Factview Budget
Factview Forecast
Factview Invoices
Factview Orders

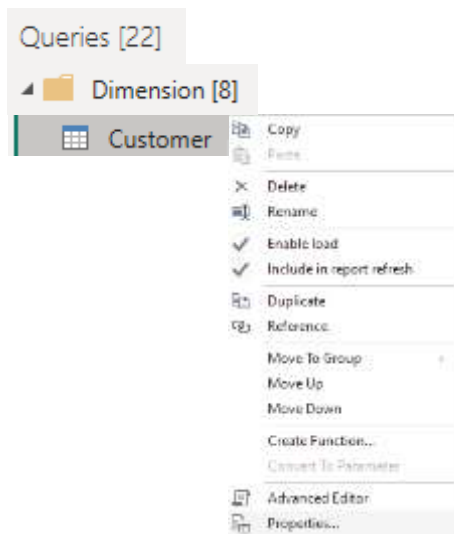


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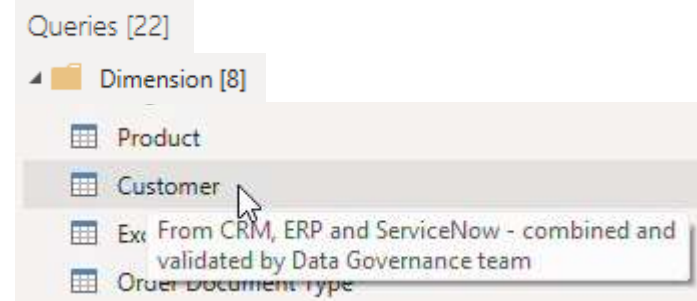
Document **table**
properties



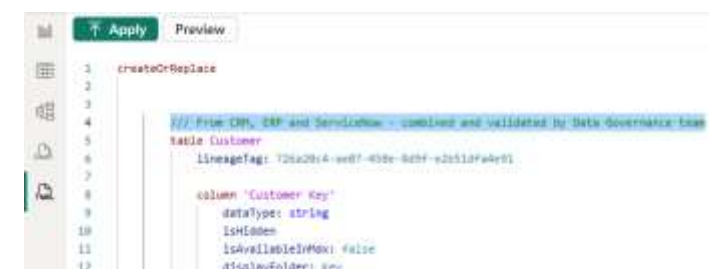
Table properties – add it once and see it everywhere



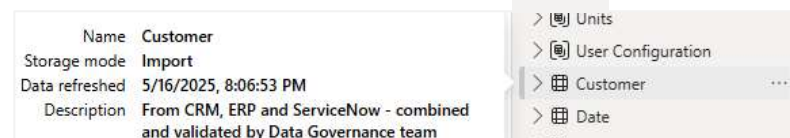
In Power Query Editor



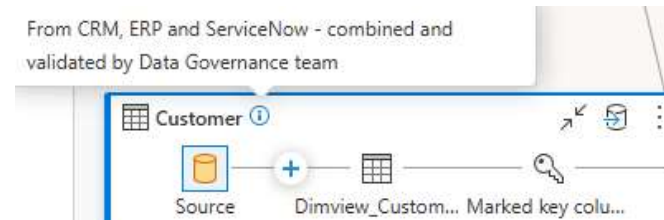
In TMDL



In Power BI Desktop



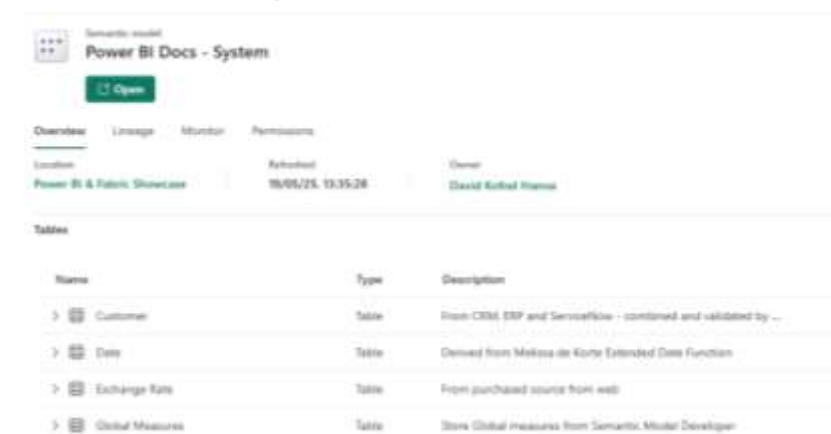
In Dataflows in the service



In INFO.VIEW DAX Functions



In OneLake catalog



Query Properties

Name
Customer

Description
From CRM, ERP and ServiceNow - combined and validated by Data Governance team

☒ Enable load to report

☒ Include in report refresh

OK Cancel

20 Power Query {M}agic Tips & Tricks

 Power BI Desktop  Fabric only

1.



Reference is preferred –
Medallion Architecture is that you?

2.



Choose Columns wisely –
Get the Gear Icon

3.



Don't create relationships
on attributes – create
surrogate keys

4.



Composite Key Merge
for multiple column join

5.



Red text will break you –
MissingField.Ignore or
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Change settings with
autodetect column types for
unstructured sources

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TMDL Script for **Date** table

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**Don't use the Excel
Connector** when connecting
to Excel – use **Web**

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Unpivot – It feels like magic

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Stop Find and Replace for
regional formats by **Using
Locale...**

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Structure your **queries** in
folders

12.



Document **table properties**

13.



From Power Query to
Fabric Dataflows Gen 2 with
copy-paste

14.



**Dynamically Reference Your
Tables** for Table.Profile

15.



**Snapshots with Append to
Lakehouse**

16.



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you use Excel and .csv you
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17.



Look for the **stream** in View
Query Plan for optimization

18.



You can now **write SQL-
Views** with Visual Query
Editor

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From **Excel** to **SQL** Analytics
Endpoint in Lakehouse

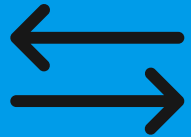
20.



Orchestration Pipelines with
invoke Dataflow gen2 &
Semantic Model refresh

{13}

From **Power Query** to
to **Fabric Dataflows Gen 2**
with **copy-paste**



{14}

**Dynamically Reference Your
Tables** for Table.Profile



There should be **one** {M}

 **Marco Russo** ✓
@marcorus

Voted!

There should be one “M”

We need more votes!

#powerbi #dataflow

Oversæt post

 **Ben Gribaudo** @bgribaudo · 22. sep. 2023

Asking for #sections to be fixed in Power BI's #dataflows – #PowerQuery
#Fabric #PowerBI

ideas.fabric.microsoft.com/ideas/idea/?id...

Fix #sections in Power Query in Dataflows

Idea Options ▾

10

 **Fix #sections in Power Query in Dataflows**
thcideas_migusr on 09-22-2023 05:41 PM ·

 Voted

Per the M language specification, #sections[Section1] should return a record containing one field per query in the project, with that field's value set to the results of the given query.

Power BI Desktop and Excel work this way. However, Power Query in dataflows **does not**. This limits the ability to write M code that dynamically adapts based on the queries present in a project.

Can PQ in dataflows be fixed so that #sections[Section1] works according to the specification?

Thank you for your consideration!

To Reproduce

In both Power BI Desktop and in dataflows, create the following two queries:

- A—defined as “123”
- B—defined as “#sections[Section1]”

In both Power BI Desktop and dataflows, the output of B will be a record that includes a field named “A”. In Desktop, this field's value will be 123 (expected behavior, per the M spec); however, in dataflows, this field's value will be null (incorrect behavior).

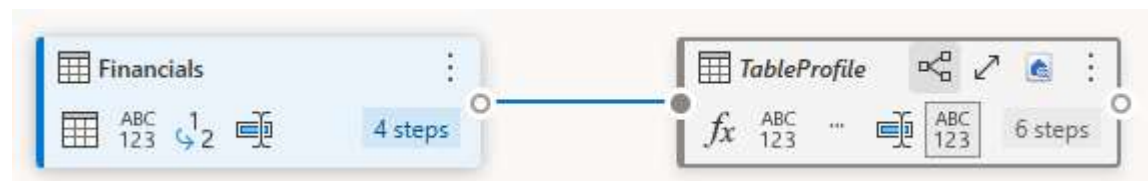
NEEDS VOTES

{15}

Snapshots with Append to Lakehouse



Data Destination & Snapshots with Append



```
1 let
2   Source = Table.Profile(Financials),
3   #"Changed column type" = Table.TransformColumnTypes(Source, {{"Count", type number}, {"NullCount", type text}, {"DistinctCount", type text}, {"Average", type text}, {"Max", type text}, {"Min", type text}}),
4   #"Added custom" = Table.AddColumn(#"Changed column type", "Table", each "Financials"),
5   #"Added custom 1" = Table.AddColumn(#"Added custom", "Custom", each DateTime.LocalNow()),
6   #"Renamed columns" = Table.RenameColumns(#"Added custom 1", {{"Custom", "DateTime"}}),
7   #"Changed column type 1" = Table.TransformColumnTypes(#"Renamed columns", {{"DateTime", type datetime}, {"Table", type text}})
8 in
9   #"Changed column type 1"
```

Tab moves focus.

	Column	Min	Max	Average	1.2 StandardDeviation	1.2 Count	NullCount	DistinctCount	Table	DateTime
1	COGS	\$1.900,00	\$90.740,00	null	null	35 0	35	35	Financials	21/08/2025, 12:32:49
2	Country	Canada	United States of America	null	null	35 0	5	5	Financials	21/08/2025, 12:32:49
3	Date	01/01/2015	01/01/2015	01/01/2015	null	35 0	1	1	Financials	21/08/2025, 12:32:49
4	Discount Band	High	None	null	null	35 0	4	4	Financials	21/08/2025, 12:32:49
5	Discounts	\$-	\$822,15	null	null	35 0	34	34	Financials	21/08/2025, 12:32:49
6	Gross Sales	\$118.375,00	\$96.600,00	null	null	35 0	35	35	Financials	21/08/2025, 12:32:49
7	Manufacturing Price	\$10,00	\$5,00	null	null	35 0	6	6	Financials	21/08/2025, 12:32:49
8	Month Name	January	January	null	null	35 0	1	1	Financials	21/08/2025, 12:32:49
9	Month Number	1	1	1	0	35 0	1	1	Financials	21/08/2025, 12:32:49
10	Product	Amarilla	Velo	null	null	35 0	6	6	Financials	21/08/2025, 12:32:49
11	Profit	\$(12.787,50)	\$7.718,40	null	null	35 0	35	35	Financials	21/08/2025, 12:32:49
12	Sale Price	\$12,00	\$7,00	null	null	35 0	7	7	Financials	21/08/2025, 12:32:49
13	Sales	\$10.733,40	\$87.906,00	null	null	35 0	31	31	Financials	21/08/2025, 12:32:49
14	Segment	Channel Partners	Small Business	null	null	35 0	5	5	Financials	21/08/2025, 12:32:49
15	Units Sold	322	2931	1445.74285714285...	859.5886274	35 0	35	35	Financials	21/08/2025, 12:32:49
16	Year	2015	2015	2015	0	35 0	1	1	Financials	21/08/2025, 12:32:49

Data Destination & Snapshots with Append

 : |

Default destination

 No default destination available

New destination

-  Lakehouse
-  Warehouse
-  SQL database
-  Azure SQL database
-  Azure Data Explorer (Kusto)
-  SharePoint
- More...

▼ Data destination

✕  Lakehouse

 Workspace

20 Power Query (M)Tips

 Lakehouse

DataProfile

 Table

TableProfile

 Update method

Append

 Column mapping

Manual mapping with fixed schema

Data destination

Choose destination target

☐ New table ☒ Existing table

Search

Display options

- Lakehouse [11]
- 20 Power Query (M)Tips [2]
- DataflowsStagingLakehou... [2]
- DataProfile
- ErrorHandling
- TableProfile
- Lakehouse
 - Digital Skill Enhancer
 - Fabric Analyst in A Day
 - Fabric Toolbox
 - Mastering Composite Mod...
 - Microsoft Fabric Capacity ...
 - My workspace
 - Power BI & Fabric Showcase
 - Power BI Advanced Visuals...
 - Semantic Link Labs
 - Tabular Editor - Sandbox

The table TableProfile in DataProfile will be used as a destination.

TableProfile

Column	Min	Max	1,2 Average	1,2 StandardDeviation	1,2 Count	1,2 NullCount	1,2 DistinctCount	ModifiedDate	Table
COGS	918	950623	140045.0187	206796.5797	24364	0	543	06/02/2025 11:56:27	Finance Transactions
Country Key	1	5	2.997373174	1.419801675	24364	0	5	06/02/2025 11:56:27	Finance Transactions
Date	01/08/2013	01/12/2014	41756	null	24364	0	58	06/02/2025 11:56:27	Finance Transactions
Discounts	0	149077.5	13287.39265	24609.25724	24364	0	513	06/02/2025 11:56:27	Finance Transactions
Gross Sales	1799	1207330	183348.1928	269913.0709	24364	0	336	06/02/2025 11:56:27	Finance Transactions
Manufacturing Price	3	260	79.22976523	102.477364	24364	0	6	06/02/2025 11:56:27	Finance Transactions
Profit	-40617.5	262200	30021.78144	49749.49729	24364	0	557	06/02/2025 11:56:27	Finance Transactions
Sale Price	7	350	118.6773106	146.2206913	24364	0	7	06/02/2025 11:56:27	Finance Transactions
Sales	1655.08	1159000	170066.8002	251374.5791	24364	0	359	06/02/2025 11:56:27	Finance Transactions
Segment Key	1	700	356.0349581	200.5843863	24364	0	700	06/02/2025 11:56:27	Finance Transactions
Units Sold	200	4482.5	1808.473206	865.7644593	24364	0	310	06/02/2025 11:56:27	Finance Transactions
Budget	1123	893963	436658.9444	257873.5417	144	0	144	06/02/2025 11:56:22	Budget
Date	01/01/2013	01/12/2014	41624	null	144	0	34	06/02/2025 11:56:22	Budget
Product	Amarillo	Velo	null	null	144	0	6	06/02/2025 11:56:22	Budget
Budget	1123	893963	436658.9444	257873.5417	144	0	144	06/02/2025 11:58:42	Budget
Date	01/01/2013	01/12/2014	41624	null	144	0	34	06/02/2025 11:58:42	Budget
Product	Amarillo	Velo	null	null	144	0	6	06/02/2025 11:58:42	Budget
COGS	918	950623	140045.0187	206796.5797	24364	0	543	06/02/2025 11:58:43	Finance Transactions
Country Key	1	5	2.997373174	1.419801675	24364	0	5	06/02/2025 11:58:43	Finance Transactions
Date	01/08/2013	01/12/2014	41756	null	24364	0	58	06/02/2025 11:58:43	Finance Transactions
Discounts	0	149077.5	13287.39265	24609.25724	24364	0	513	06/02/2025 11:58:43	Finance Transactions
Gross Sales	1799	1207330	183348.1928	269913.0709	24364	0	336	06/02/2025 11:58:43	Finance Transactions
Manufacturing Price	3	260	79.22976523	102.477364	24364	0	6	06/02/2025 11:58:43	Finance Transactions
Profit	-40617.5	262200	30021.78144	49749.49729	24364	0	557	06/02/2025 11:58:43	Finance Transactions

Data Destination & Snapshots with Append

Data destination

Choose destination settings

To improve the performance of the data load into the destination, we are going to disable staging for the source query.

Use automatic settings

Update method

Existing data

New data

→

Append

Replace

Column mapping

☒	Source	Source type	Destination	Destination type
☒	Table	All Text	Table	Text
☒	Column	All Text	Column	Text
☒	SnapshotDate	Date/Time	SnapshotDate	Date/Time
☒	Min	All Text	Min	Text
☒	Max	All Text	Max	Text
☒	Average	All Text	Average	Text
☒	StandardDeviation	1,2 Decimal number	StandardDeviation	Decimal number
☒	Count	Whole number	Count	Whole number
☒	NullCount	Whole number	NullCount	Whole number
☒	DistinctCount	Whole number	DistinctCount	Whole number

Data Destination & Snapshots with Append

DataProfile

Home

Get data

New semantic model

Open notebook

Manage OneLake data access (preview)

A SQL analytics endpoint for SQL querying and a default Power BI semantic model for reporting were created with this item.

Explorer

Search tables

DataProfile

Tables

TableProfile

Files

TableProfile

	AKO Column	AKO Min	AKO Max	12 Average	12 StandardDeviation	12 Count	12 NullCount	12 DistinctCount	ModifiedDate	AKO Table
1	COGS	918	950625	140045.018716139	208790.575671081	24364	0	545	2/6/2025 11:56:27 AM	Finance Transactions
2	Country Key	1	5	2.99737317353472	1.41980167520308	24364	0	5	2/6/2025 11:56:27 AM	Finance Transactions
3	Date	01/09/2013	01/12/2014	41758		24364	0	16	2/6/2025 11:56:27 AM	Finance Transactions
4	Discounts	0	149677.5	13281.3926537106	24609.2572369132	24364	0	515	2/6/2025 11:56:27 AM	Finance Transactions
5	Gross Sales	1799	1207500	183346.192804958	269915.070870866	24364	0	550	2/6/2025 11:56:27 AM	Finance Transactions
6	Manufacturing Price	3	260	79.2297652273847	102.477364023807	24364	0	6	2/6/2025 11:56:27 AM	Finance Transactions
7	Profit	-40617.5	262200	30021.7814351092	49149.4973932004	24364	0	557	2/6/2025 11:56:27 AM	Finance Transactions
8	Sale Price	7	350	116.677310786406	146.22669149658	24364	0	7	2/6/2025 11:56:27 AM	Finance Transactions
9	Sales	1655.08	1158200	170086.800151245	251374.579127228	24364	0	559	2/6/2025 11:56:27 AM	Finance Transactions
10	Segment Key	1	700	356.054958134953	200.584385266284	24364	0	700	2/6/2025 11:56:27 AM	Finance Transactions
11	Units Sold	200	4492.5	1608.42320637005	868.784459334978	24364	0	510	2/6/2025 11:56:27 AM	Finance Transactions
12	Budget	1123	893965	436688.944444444	257073.941746999	144	0	144	2/6/2025 11:58:42 AM	Budget
13	Date	01/01/2013	01/12/2014	41624		144	0	24	2/6/2025 11:58:42 AM	Budget
14	Product	Amarillo	Velo			144	0	6	2/6/2025 11:58:42 AM	Budget
15	Budget	1123	893965	436688.944444444	257073.941746999	144	0	144	2/6/2025 11:56:22 AM	Budget
16	Date	01/01/2013	01/12/2014	41624		144	0	24	2/6/2025 11:56:22 AM	Budget
17	Product	Amarillo	Velo			144	0	6	2/6/2025 11:56:22 AM	Budget
18	COGS	918	950625	140045.018716139	208790.575671081	24364	0	545	2/6/2025 11:58:43 AM	Finance Transactions
19	Country Key	1	5	2.99737317353472	1.41980167520309	24364	0	5	2/6/2025 11:58:43 AM	Finance Transactions
20	Date	01/09/2013	01/12/2014	41758		24364	0	16	2/6/2025 11:58:43 AM	Finance Transactions
21	Discounts	0	149677.5	13281.3926537105	24609.2572369131	24364	0	515	2/6/2025 11:58:43 AM	Finance Transactions
22	Gross Sales	1799	1207500	183346.192804958	269915.070870867	24364	0	550	2/6/2025 11:58:43 AM	Finance Transactions
23	Manufacturing Price	3	260	79.2297652273847	102.477364023807	24364	0	6	2/6/2025 11:58:43 AM	Finance Transactions
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25	Sale Price	7	350	116.677310786406	146.22669149658	24364	0	7	2/6/2025 11:58:43 AM	Finance Transactions
26	Sales	1655.08	1158200	170086.800151245	251374.57912723	24364	0	559	2/6/2025 11:58:43 AM	Finance Transactions
27	Segment Key	1	700	356.054958134953	200.584385266283	24364	0	700	2/6/2025 11:58:43 AM	Finance Transactions
28	Units Sold	200	4492.5	1608.42320637005	868.784459334981	24364	0	510	2/6/2025 11:58:43 AM	Finance Transactions

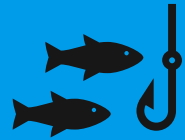
{16}

Don't Break the Fold –
but if you use Excel and
.csv you are!



{17}

Look for the **stream**



{18}

You can now **write SQL-Views**
with Visual Query Editor



{19}

From Excel to **SQL Analytics**
Endpoint in Lakehouse



{20}

Orchestration Pipelines

with invoke Dataflow Gen2
& Semantic Model refresh



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Orchestration Pipelines with
invoke Dataflow gen2 &
Semantic Model refresh

You just have a self-
service solution...

Really?

Star schema

Query Folding

Surrogate Key,
Primary Key
and Foreign
Key

Orchestration
pipeline

Split model
and report

Aggregations &
Write SQL
Views

Snapshot

“Medallion
Architecture”

Error handling

Delta Tables
Lakehouse

Composite Key
Merge

Documentation

What about **user adoption** and **solving business problems**?

Documentation

Query Folding

Orchestration
pipeline

Error handling

Snapshot

Split model
and report

Composite Key
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Surrogate Key,
Primary Key
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Aggregations &
Write SQL
Views

“Medallion
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Thank you!